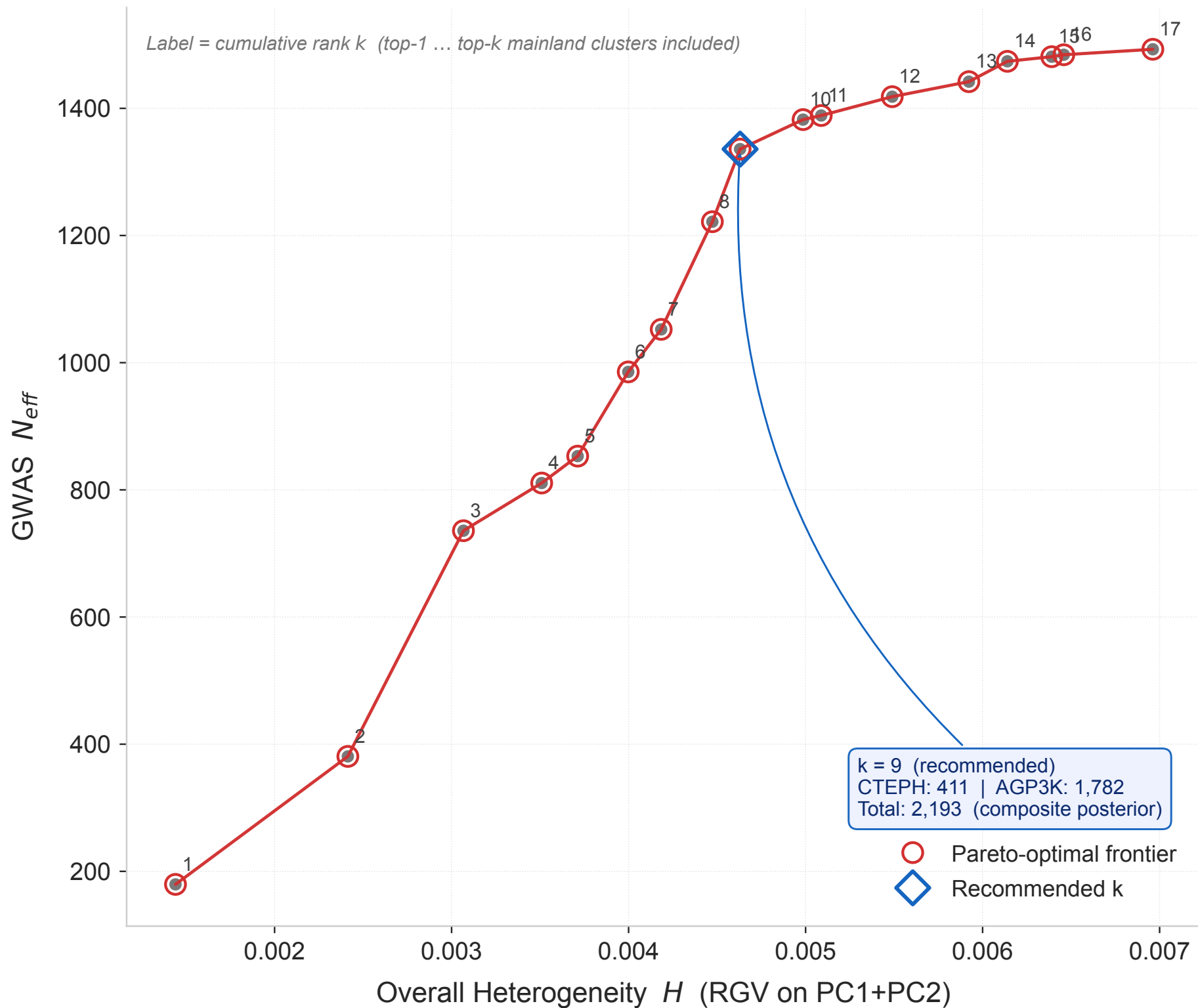


# Mainland Rank-Cumulative Trade-off Analysis

## Trade-off: Overall Heterogeneity vs. GWAS $N_{\text{eff}}$



## Effective Sample Size $N_{\text{eff}}$

Variance of allele-frequency difference (unbalanced design):

$$\text{Var} = p(1-p) \left( \frac{1}{N_{\text{case}}} + \frac{1}{N_{\text{ctrl}}} \right)$$

Equate to balanced design ( $N_{\text{eff}}/2$  per arm):

$$= \frac{2p(1-p)}{N_{\text{eff}}}$$

$$\Rightarrow N_{\text{eff}} = \frac{4N_{\text{case}}N_{\text{ctrl}}}{N_{\text{case}} + N_{\text{ctrl}}} = N_{\text{tot}} \cdot \frac{4r}{(1+r)^2}$$

Equivalent total  $N$  of a balanced (1:1) case-control study with the same statistical test power.

( $r = N_{\text{case}}/N_{\text{ctrl}}$ ,  $N_{\text{tot}} = N_{\text{case}} + N_{\text{ctrl}}$ )

## Overall Heterogeneity $H$

$$H = |\Sigma_{PC1, PC2}|^{1/4}$$

$$|\Sigma| = \sigma_{PC1}^2 \cdot \sigma_{PC2}^2 - \sigma_{PC1, PC2}^2 = \sigma_{PC1}^2 \cdot \sigma_{PC2}^2 (1 - \rho^2)$$

$$\Rightarrow H = (\sigma_{PC1} \cdot \sigma_{PC2})^{1/2} (1 - \rho^2)^{1/4}$$

Root Generalized Variance (2D): joint spread in PC1/PC2, attenuated by inter-PC correlation  $\rho$ .

## Counting methods:

Rank table: per-component argmax (MAP), same as STEP4 Panel D.

Scatter plot: composite posterior recomputation (top- $k$  merged), same as STEP5.